Definition

$$v = \frac{x_2 - x_1}{t_2 - t_1} = \frac{\Delta x}{\Delta t}$$

Angular Definitions

$$\omega = \frac{\theta_2 - \theta_1}{t_2 - t_1}$$

Circular Motion

$$v = \omega r$$

Momentum

$$p = mv$$

$$m_1 v_1 = m_2 v_2$$

$$a = \frac{v_2 - v_1}{t_2 - t_1} = \frac{\Delta v}{\Delta t}$$

$$\alpha = \frac{\omega_2 - \omega_1}{t_2 - t_1}$$

$$a_c = \omega^2 r = \frac{v^2}{r}$$

a_c aka a_{radial}

Stick

$$m_1 v_1 + m_2 v_2 = (m_1 + m_2) v_f$$

A Relative B (Sign Matters)

$$v_{A/B} = -v_{B/A}$$

$$v_{A/A} = v_{A/B} + v_{B/A}$$

$$s = r\theta$$

$$F_C = \frac{mv^2}{r} = m\omega^2 r$$

Energy

$$W = Fx\ (\cos\varphi)$$

$$W = \frac{1}{2}kx^2$$

cons of energy

SUVAT

$$v = u + at$$

no x

Angular

$$\omega = \omega_0 + \alpha t$$

where α = rad s^{-2}

$$\omega = \frac{2\pi}{T}$$

angular velocity definition

$$K_E = \frac{1}{2}mv^2$$

$$KE_i + PE_i = KE_f + PE_f$$

energy momentum

$$x = x_0 + ut + \frac{1}{2}at^2$$

no v

θ is angular displacement

$$\theta = \omega t + \frac{1}{2}\alpha t^2$$

θ can be rev / rad -> ω follows

$$a_{tan} = r\alpha$$

$$GPE = mgh$$

Motion ↑ ⇒ GPE $-ve \Rightarrow$ *as* $g\downarrow$

Mechanical Energy = KE + GPE + EPE

$$x = x_0 + \frac{1}{2}(u + v)\,t$$

no a

$$\theta = \frac{1}{2}(\omega + \omega_0)t$$

$$v_{max} = \sqrt{\mu_s gR}$$

flat road

Friction on a Slope

$$W_{net} = -\mu_k N x$$

F x

$$v^2 = u^2 + 2a\,\Delta x$$

no t

$$\omega^2 = \omega_0^2 + 2\alpha\theta$$

$$tan\beta = \frac{v^2}{gR}$$

banked road

Power

$$P = \frac{W}{t} = Fv$$

Projectile Shortcut

$$T = \frac{2u\,sin\theta}{g}$$

time to hit ground

Newton's Second Law

$$F_{net} = ma = \frac{m\Delta v}{t}$$

Moment of Inertia

$$K_{tot} = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

ω can be $\frac{v}{r}$

Gravitation

$$F_G = \frac{GMm}{r^2}$$

$$H = \frac{u^2 (sin\theta)^2}{2g}$$

$$F_{net} = \frac{p}{t} = \frac{mv - mu}{t}$$

$$K_{Total} = K_{Trans} + K_{Rot}$$

$$K_{rot} = \frac{1}{2}I\omega^2$$

$$g = \frac{GM}{r^2}$$

$$R = \frac{u^2\,sin2\theta}{g}$$

max range

$$w = mg$$

Thin Cylindrical Shape I

$$I = mr^2$$

$$T^2 = \frac{4\pi^2 r^3}{GM}$$

Kepler's 3rd Law

$$y = x\,tan\theta - \frac{gx^2}{2u^2 (cos\theta)^2}$$

$$F_R = \sqrt{F_x^2 + F_y^2}$$

$$\theta = tan^{-1}\left(\frac{F_y}{F_x}\right)$$

Torque / Moment

$$\tau = Fl$$

perpendicular distance

$$v = \sqrt{\frac{GM}{R}} = \sqrt{\frac{GM_{earth}}{R_{earth} + R}}$$

Static and Kinetic Friction

$$f_s \leq \mu_s N$$

$$f_k = \mu_k N \quad \mu_k < \mu_s$$

Impulse

$$J = \Delta p = mv_f - mv_i$$

short period of time

Rolling without Slipping CoM

$$v_{cm} = r\omega$$

$$V_g = -\frac{GM}{r}$$

Spring

$$F = -kx$$

CoM

$$x = \frac{m_1 x_1 + m_2 x_2 + m_3 x_3}{m_1 + m_2 + m}$$

x can also be v

$$a_{cm} = r\alpha$$

$$PE = -\frac{GMm}{r}$$

Consider Energy when Stuck!

$F_D = Dv_T^2$

Drag Force = Constant x Terminal Velocity

$Dv^2 = mg$

Drag Force = Free Fall

$a = \frac{v^2}{r} \Rightarrow v = \sqrt{gL}$

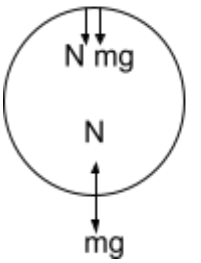
special case

Density

$\rho = \frac{m}{V}$

$\rho_{xi} = \rho_{xf}$ consider

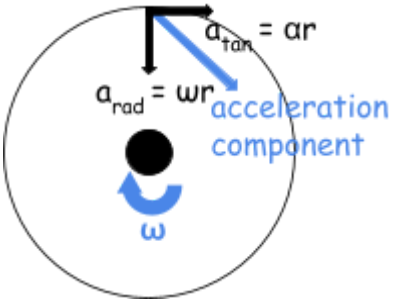
$\rho_{yi} = \rho_{yf}$ separately



$N + mg = F_C$

where $F_C = mr\omega^2$

$N - mg = F_C$



$G = 6.67 \times 10^{-11}\ N\ m^2\ kg^{-2}$

$M_E = 5.98 \times 10^{24}\ kg$

$R_E = 6.38 \times 10^{6}\ m$

Elastic Collision
- KE+ρ – Conserved

Inelastic Collision
- KE Not Conserved
  Converted into PE

$Degree = Radian \times \frac{180^o}{\pi}$

$Radian = Degree \times \frac{\pi}{180^o}$

$rad/s \times \frac{60}{2\pi} = rpm$

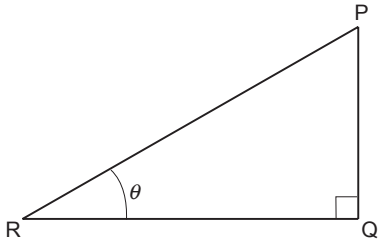
$rpm \times \frac{2\pi}{60} = rad/s$

$T = \frac{60}{rpm}$

$T = \frac{2\pi}{\omega}$

$Horsepower = \frac{Watts}{746}$

$Radians = 2\pi \times Revolutions$

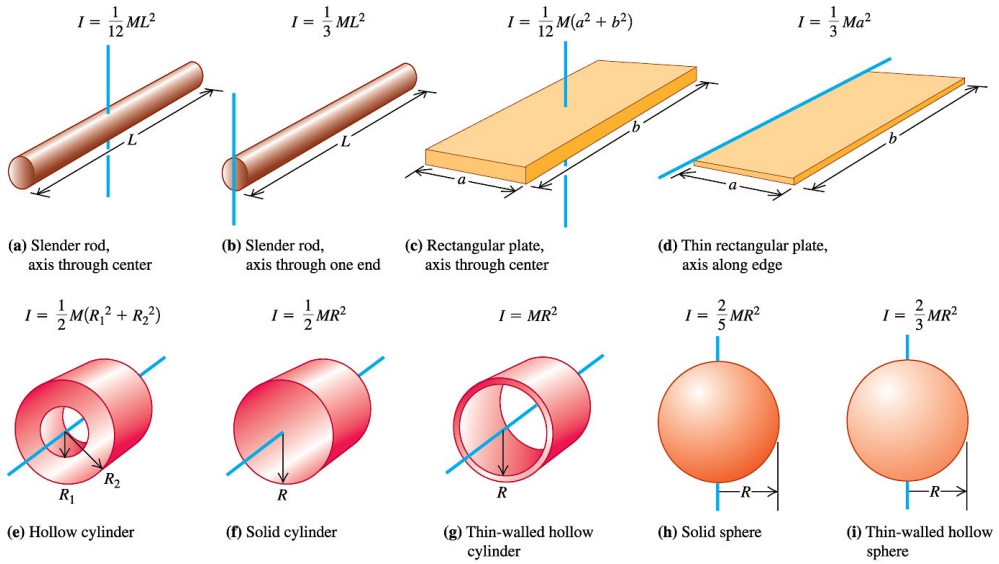


$\sin\theta = \frac{PQ}{PR}$, $\cos\theta = \frac{QR}{PR}$, $\tan\theta = \frac{PQ}{QR}$, $\frac{\sin\theta}{\cos\theta} = \tan\theta$

$PR^2 = PQ^2 + QR^2$

Shape	Surface area	Volume
circle	$\pi r^2 = \frac{\pi d^2}{4}$	—
cylinder	$2\pi r\,(r + h)$	$\pi r^2 h$
sphere	$4\pi r^2$	$\frac{4}{3}\pi r^3$

TABLE 9.2 Moments of inertia for various bodies



Multiple	Prefix	Symbol
10^{-12}	pico	p
10^{-9}	nano	n
10^{-6}	micro	μ
10^{-3}	milli	m
10^{-2}	centi	c
10^{3}	kilo	k
10^{6}	mega	M
10^{9}	giga	G